



STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSIT
Course Outcome	5

Case Study Topic: City Delivery Network – Optimal "Find Shortest Delivery Route & Maximize Truck Payload" using Graph (Non-Linear DS), Dijkstra's Greedy Algorithm, and 0/1 Knapsack DP

Word Done Summary:

In this case study, common practical applications of Non-Linear Data Structures, Greedy Algorithms, and Dynamic Programming were designed, developed, and evaluated using a city delivery network. Delivery hubs and the roads connecting them were modeled as a weighted Graph — a Non-Linear Data Structure — stored as an adjacency list, with a min-heap (priority queue) used to repeatedly track the next-nearest unvisited hub. Dijkstra's Greedy Algorithm was applied to compute the shortest route from the depot to every hub by always extending the locally nearest hub first, which is provably optimal here because the shortest-path problem has both optimal substructure and the greedy-choice property for non-negative edge weights. Once the shortest distances were known, a 0/1 Knapsack Dynamic Programming solution was applied to decide which indivisible delivery packages to load onto a capacity-limited truck so as to maximize total delivery value. Dynamic Programming was required instead of Greedy for this second problem because, unlike shortest-path routing, the 0/1 Knapsack problem does not satisfy the greedy-choice property when items are indivisible.

Implementation Details:

1. Graph construction using an adjacency list (Non-Linear Data Structure) to represent delivery hubs and road distances.
2. Min-heap (priority queue) implementation to efficiently extract the next-nearest unvisited hub.
3. Dijkstra's Greedy Algorithm to compute the shortest distance from the depot to every hub.
4. Greedy-choice and optimal-substructure justification for Dijkstra's correctness on non-negative weights.
5. 0/1 Knapsack Dynamic Programming table $dp[i][w]$ construction for truck-payload optimization.
6. Backtracking through the DP table to identify which packages are loaded onto the truck.
7. Complexity analysis: $O((V+E) \log V)$ for Dijkstra with a min-heap, $O(n \cdot W)$ for 0/1 Knapsack DP.

8. Verification of shortest distances and optimal payload value against expected results.

Result:

The Graph was constructed correctly with each hub and road weight stored in the adjacency list, and Dijkstra's Greedy Algorithm correctly computed the shortest distance from the depot to all hubs using the min-heap. The 0/1 Knapsack DP table was built correctly, and backtracking through it identified the optimal subset of packages, confirming the maximum achievable delivery value within the truck's weight capacity.

Output:

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Graph (Non-Linear DS) – Shortest Distances from Depot (Dijkstra, Greedy):  
Depot=0km  HubB=3km  HubC=1km  HubD=4km  HubE=7km
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0/1 Knapsack (DP) – Truck Capacity = 15 kg:  
Packages available: P1(12kg,60) P2(7kg,50) P3(5kg,30) P4(3kg,20)  
Packages selected : P2 + P3 + P4  (weight = 15 kg)
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Maximum Delivery Value: 100
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GitHub Link:

<https://github.com/vigneshkumar-stack/DSA-Project>

Student Signature	Faculty Signature
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